

Smart Attendance Automation System

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Abstract. The smart attendance marking system, which is based on face recognition technology, is a modern solution that may aid in the automation of the attendance management process in a variety of organizations, including schools, colleges, and corporations. Using facial recognition technology, the research team hopes to create an efficient system for tracking and recording attendance. The system will be developed by gathering and analysing data in order to construct a database of faces that will be used to train deep learning algorithms to recognise faces effectively. Compared to traditional attendance tracking systems, the smart attendance marking system based on face recognition technology has various advantages. For starters, it saves time and lowers manual labour because attendance is automatically tracked. Second, it enhances precision, lowering the possibility of errors. Third, it delivers real-time attendance updates, making it simple for instructors or administration to monitor attendance and take appropriate action. The aim of the work is expected to improve attendance management in various organisations by optimising attendance monitoring methods while decreasing manual effort and errors. Because the system may be connected with existing attendance management systems or utilised as a standalone solution, it is a versatile and adaptable option for a variety of organisations. The paper's effort is to design, develop, and implement an efficient system that can track and record attendance accurately using facial recognition technology, giving benefits such as time savings, accuracy, real-time updates, and better security.

1 Introduction

Many organisations, including schools, colleges, and corporations, require attendance tracking. Attendance has traditionally been recorded manually, which may be a time-consuming and error-prone process. However, as technology has advanced, smart attendance marking systems that can automate the attendance monitoring process have evolved, saving time and lowering the chance of errors. Face recognition technology is used in one such smart attendance marking system. This technology analyses individuals' facial traits and matches them with faces saved in a database using powerful algorithms and artificial intelligence. When a match is found, the attendance is automatically marked. There are various processes involved in developing a smart attendance marking system

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based on face recognition technology. To begin, data is collected and analysed using a facial recognition device, such as a camera, to generate a database of faces. This database is then used to train deep learning algorithms to reliably recognise faces. With the addition of new data, the algorithms are constantly refined, which helps to enhance accuracy over time. Once constructed, the system can be linked into existing attendance management systems or used as a stand-alone solution. The system can be accessible via a web-based or mobile application, allowing students or staff to mark their attendance from any location. Furthermore, the system can provide real-time attendance updates, making it simple for faculty or administration to monitor attendance and take appropriate action. A smart attendance marking system based on face recognition technology has various advantages over traditional methods of attendance tracking. For starters, it saves time and lowers manual labour because attendance is automatically tracked. Second, it enhances precision, lowering the possibility of errors. Third, it delivers real-time attendance updates, making it simple for instructors or administration to monitor attendance and take appropriate action. Finally, it improves security by preventing unauthorised entry to critical places by only allowing access to persons whose faces the system recognises.

2 Literature survey

Dhanush Gowda and team members adopted Convolutional Neural Network (CNN). The specific techniques used are Image processing, Face detection, Feature matching, Face Recognition. The Accuracy for this is 98%. A hybrid model combining CNN and Dlib achieved better results compared to using either method individually. Time taken for matching the 128 encodings of each frame with each picture in the database is time consuming. The authors [2] adopted face recognition under uncontrolled conditions. The specific technique used is the system takes an image as input and allows only one person per image. The accuracy for this model is 97%. The accuracy is fairly high as there is only a single person per image in the input. Does not meet the real time requirements. Winarno and his team [3] adopted dual vision face recognition using 3 WPCA. The specific Requirements are Pictures from dual vision cameras are combined using half join method and 3WCPA extraction is applied. The accuracy for this model is 89%. This Face recognition System will predict the fake faces from the real ones, If any photographs are shown it'll consider them as fake and only identify the real human faces and takes them into consideration. Raveen [4] implemented Intelligent Attendance System based on facial Identification. The methodology used for this Model is ADABOOST algorithm with processes like Principal Component Analysis and LBP Hybrid algorithm which is used in most face recognized systems. The specific techniques are Attribute reduction, Statistical and Unsupervised classification algorithms are used. The accuracy for this model is 97%. With this usage of this kind of marking system, The attendance will be more proxyless and more accurate, it will not recognise the fake persons from photographs and only marks attendance for the real ones. Nusrat Mubin Ara1 [5] adopted Alex NET CNNs and RFID Technology. The specific Techniques used is Neural networks are used for face detection and recognition paired with Radio Frequency based devices. The accuracy for this model is 96%. This model uses a camera as the primary resource for capturing. This won't be applied until it is confirmed that it is fully working on students. RFID technology uses electronic toys which can't be used in all cases.

Siti Umami Masruroh [6] suggested iris recognition for authentication. The specific techniques used is Iris Recognition is used to identify a person using neural networks. The Accuracy for this model is 98%. Real time face detection and efficient Poor light conditions lead to inaccurate identification of a person. Radhika [7] suggested face recognition-based attendance system using machine learning algorithms. The methodology used is SVM +

LDA. The specific techniques used are Deep learning Algorithms with some specialised approaches with more improved accuracy. The accuracy is 95%. Regular and spontaneous attendance system. Model is not performing as well as algorithms work. Wenxian Zeng [8] depicts model based on deep learning and face recognition. The methodology used is CNN. The specific techniques used in this model is Convolutional Neural Network uses a multilayer perceptron that works faster and great with good accuracy. The accuracy for this model is 98%. Helps to achieve high-precision real-time attendance with increased speed and accuracy to satisfy the demand for automatic classroom evaluation. There are other, more precise construction techniques available. Akshith and their team [9] used automated marking System Using Deep learning. The methodology used for this model is DWT. The specific requirements are Grayscale standardisation, histogram balance, Discrete Wavelet Transform (DWT) faster. The accuracy for this model is 82%. Regarded as the highest recognition rate attainable from the data. It is possible to get greater accuracy.

Chandramouli [10] followed attendance marking System using histogram plots. The methodology used in this model are Haar Classifier and LBPH algorithm. Faces are detected by deep learning algorithms which use a HAAR classifier for classifying faces , next step is recognized by Local Binary pattern Histogram algorithm, the acquired graphical results are verified against the given database and attendance will be marked. The accuracy is 93%. One of the innovative methods and algorithms used to perform the attendance marking with high accuracy. Works slow under some circumstances and problems occur during histogram plotting. Susanto [11] suggested Android and face recognition application. The methodology used in this model are OpenCV and LBPH algorithms. Making an Android application by utilising OpenCV for speed detection. Facial recognition will use the LBPH Histogram. This model has an accuracy rate of 89%. regarded as the highest recognition rate attainable from the data. It is possible to have greater precision. Kumar [12] adopted methodology used in this model is GPS location. Students' whereabouts are determined using GPS on their phones. This is known as the key to recording attendance.. Accuracy for this model is 70%. Doesn't require a student's face to mark attendance, Using his location attendance can be marked. Always a mobile device with him, prone to faulty attendance.

Prangchumpol [13] implemented face recognition-based attendance system is the name of the paper. CNN is the approach employed in this model. CNN employs a system similar to a multilayer perceptron that is built to handle the requirements more quickly. The model's accuracy is 98%.Uses the camera system to keep track of scene data. There are other, more precise building techniques available. Singhal and Gujral [14] implemented RFID remote monitoring attendance system by sending an SMS approach using GSM Module. The specific requirements are RFID.The accuracy for this model is 85%. Highly used in the academic sector for student monitoring purposes. This model is highly accurate, Error rate is very low. Students can cheat easily by using their images on mobile phones for their actual face. PVN and Gupta [15] implemented Smart attendance using Fingerprint. The methodology used is the Fingerprint and GSM module. Conducted a portable attendance system by using a GSM network with fingerprint technique to manage and record. The accuracy for this model is 96%. Authors [16] highlighted the significance of ML in prediction, pattern recognition and error reduction across diverse fields, emphasizing the impact of AI in broad domain. Authors [17] discussed about the design of computerized certified vehicle identification device that makes use of a vehicle number plate. Authors developed a system that identifies some of the atomic actions from images [18]. Authors [19] developed a web based application for helping people to handle their emotions without any external help.

3 Problem statement and objectives

3.1 Problem statement

The objective of the paper is to design and develop a smart attendance automated system that utilises facial recognition technology. The system will be designed to recognize faces in real-time, mark attendance automatically, and provide real-time updates. To build an effective facial recognition system, a large collection of faces must be collected. The paper will entail gathering and analysing data in order to build a database of faces that will be used to train deep learning algorithms to recognise faces effectively. Deep learning techniques will be used by the system to reliably recognise faces. Then the recognized faces will be added into the attendance sheet.

3.2 Objective of the paper

- To capture the student's faces and store it.
- Recognizing the faces of the students with the help of trained model.
- To provide attendance for the students who are present in the class and subsequently sending the absentees list to concerned faculty through email.

3.3 Architecture of the proposed work

The smart attendance automated system based on facial recognition architectural diagram is made up of numerous critical components that work together to build a comprehensive system. The system starts with an input layer that captures photos of faces, which are subsequently processed and prepared by a pre-processing layer. The pre-processing layer may contain algorithms for resizing photos, normalising lighting and colours, and extracting facial traits for recognition. When the photos are ready, they are forwarded to the facial recognition layer for analysis. The system's facial recognition layer is in charge of accurately recognising faces. This layer often incorporates deep learning algorithms that have been trained on a collection of faces to reliably recognise distinct persons. The facial recognition layer may entail fine-tuning the models to improve accuracy and testing the models to verify proper operation. Once a face is detected, the system's attendance management layer is in charge of maintaining attendance records. This layer may include actions like attendance marking, updating attendance records, and creating reports. This layer may include responsibilities like database design, database maintenance, data backup and recovery, and so on. Finally, the system's security layer is in charge of ensuring the system's security and preventing unauthorised access. Face recognition-based authentication, user access control, and data encryption are all possible tasks for this layer. Overall, the architecture diagram for the smart attendance marking system based on facial recognition is intended to ensure that all components operate flawlessly together to give an accurate, dependable, and secure attendance marking solution.

3.4 Modules

3.4.1. Camera module

The camera module is a critical component of the facial recognition-based smart attendance marking system. It takes pictures of pupils or staff and sends them to be pre-processed and recognised. The camera module in this system should be of good quality, with a high

resolution, and capable of capturing images in a variety of lighting circumstances. To capture clear and sharp photographs of moving humans, the camera module should include capabilities such as auto-focus, auto-exposure, and a fast shutter speed. The camera should also feature a wide-angle lens so that numerous faces can be captured in a single frame, increasing efficiency and decreasing waiting times during the attendance marking process. Furthermore, the camera module should be integrated into the system and interface with other modules such as the pre-processing and identification modules seamlessly. With this integration, the camera module will be able to capture images and transfer them for further processing without the need for manual involvement.

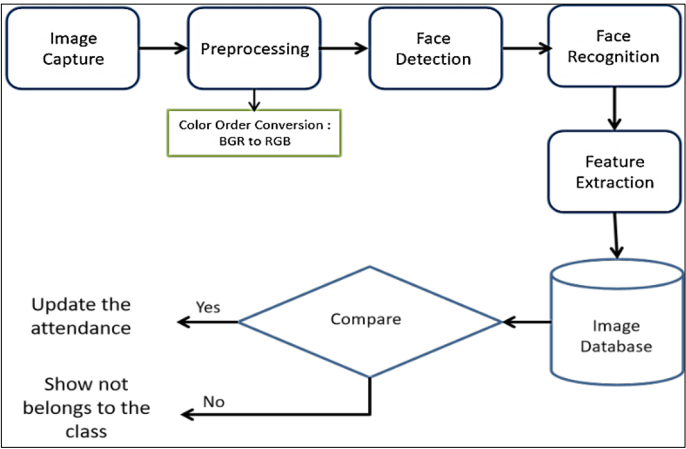


Fig. 1. Architecture diagram.

3.4.2. Data pre-processing module

The pre-processing module is critical in any facial recognition system since it prepares the input photos for the recognition phase. The pre-processing module in the smart attendance automated system based on face recognition is in charge of numerous functions, including resizing photos, normalising lighting and colours, and extracting facial traits for recognition. The pre-processing module's principal job is to shrink the incoming photos to a standard size. Because facial recognition algorithms often perform better with standardised sizes, photos must be resized. It also assures that the system uses the same size for all photographs, eliminating the possibility of distortions during the recognition process. Normalising the lighting and colours of the input photos is another key job of the pre-processing module. This assignment is critical since the lighting and colours in the photographs can have a substantial impact on facial recognition accuracy. The normalisation procedure entails altering the photos' brightness, contrast, and colour balance to guarantee that the system can reliably recognise faces. The pre-processing module is also in charge of extracting facial traits for recognition, in addition to resizing and normalising. This procedure entails identifying and extracting facial features such as the eyes, nose, mouth, and other facial landmarks. Once these features are identified, they are analysed and matched to those in the face database in order to identify the person.

3.4.3. Face recognition module

The face recognition module is a critical component of the facial recognition-based smart attendance automated system. This module is in charge of identifying and recognising faces

in input photos and comparing them to stored face templates. The module use deep learning techniques to extract and analyse an individual's face traits, such as the shape of the nose, mouth, and eyes, as well as the distance between these features. A machine learning model that has been trained on a large dataset of faces is often included in the face recognition module. This model use a technique known as Convolutional Neural Networks (CNNs) to extract features from input photos and compare them to previously stored face templates. Several pre-processing approaches are utilised to improve the accuracy of the face recognition module. Image normalisation, in which the input image is modified for brightness, contrast, and colour balance, and image alignment, in which the face is aligned to a standard location, are two of these procedures. These strategies ensure that facial

The face recognition module may also perform facial verification, which involves comparing a person's identity to a database of known identities. This is useful when the system has to authenticate a person's identification before giving entry to a restricted location or executing a financial transaction.

3.4.4. GSM module

The GSM (Global System for Mobile Communications) module is an essential component of the facial recognition-based smart attendance automated system. The module is in charge of delivering notifications to parents' or guardians' registered mobile numbers when their child enters or quits the school grounds. This assures the pupils' safety and security, and their locations can be easily followed by their parents or guardians. The module is a small device that talks with the mobile network operator (MNO) via a SIM card and is integrated into the attendance system. When a student's attendance is marked by the facial recognition system, the module sends an SMS notice to the student's parent or guardian's registered cell phone number. This module runs in a low-power mode, allowing it to run on a backup battery in the case of a power outage. It also has a real-time clock, allowing it to correctly track the current time. This feature guarantees that SMS notifications sent to parents or guardians include the right date and time, giving them up-to-current information about their child's attendance. The GSM module is a versatile component that may be configured to send notifications to many phone numbers at the same time. This tool is useful when parents or guardians cannot be reached at their primary phone numbers. Furthermore, the module can be set to deliver messages just when a student is absent or fails to check in. This setting ensures that only necessary messages are sent to parents or guardians. The integration of the GSM module in the smart attendance marking system based on facial recognition means that parents or guardians are kept up to date on their child's attendance in real time, giving them peace of mind and safeguarding the students' safety and security.

4 Results and discussions

4.1 Description of the dataset

The dataset refers to the collection of photographs of faces that will be used for training and testing the facial recognition model in the context of the paper. The dataset is critical to the system's performance because the accuracy of the facial recognition model is dependent on the quality and number of photographs in the dataset. This paper's dataset must be broad and diverse, comprising photographs of people of various ethnicities, genders, ages, and lighting situations. The photographs should also include a variety of facial expressions, perspectives, and stance changes. The dataset should be representative of the system's target

population and contain a wide range of potential scenarios that the system may meet in real-world circumstances. A high-quality dataset is required to assure the accuracy and reliability of the facial recognition model. This means that the photographs in the dataset should have a high enough resolution and clarity to capture the facial traits required for accurate recognition. The photos should also be well-labelled, with each image representing a unique individual in the database. Labelling should be precise and consistent, with no mistakes or misidentifications. Aside from image quality, the dataset should be well-organised and conveniently available for training and testing the facial recognition algorithm. The dataset should be saved in an organised format to enable for efficient image retrieval and processing. Finally, ensure that the dataset is acquired and handled in accordance with ethical and legal criteria. This includes gaining informed consent from individuals whose photographs are included in the dataset, as well as protecting their privacy and security. It also entails ensuring that the dataset is not used to discriminate against individuals or groups based on protected characteristics such as race, gender, or ethnicity.

4.2 Experimental results

The result obtained after recognition of faces using face recognition module in python, which is an inbuilt module that provides neural network models for training with faces and providing us with the facial features



Fig. 2. Detecting faces from motion picture.

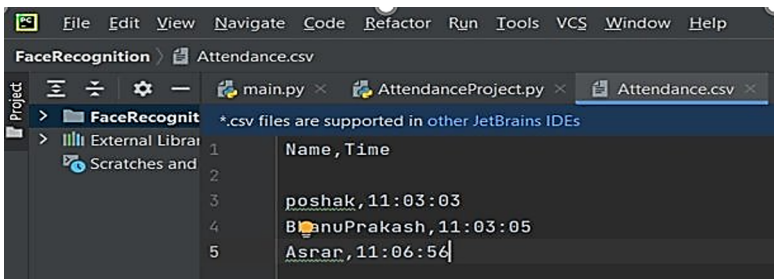


Fig. 3. Marking attendance for the recognized faces.



Fig. 4. Recognizing faces from a class.

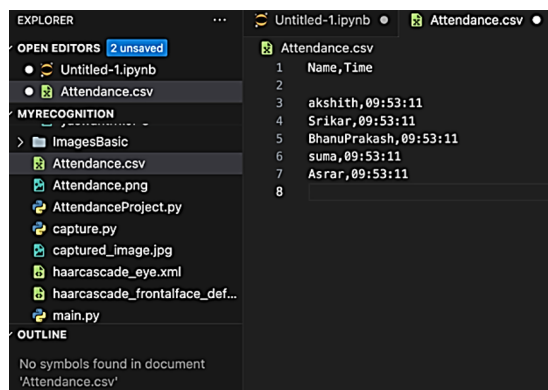


Fig. 5. Marking attendance for the recognized students.

5 Conclusion and future enhancements

It's challenging to give a detailed response without understanding the particulars of the suggested strategy. However, in comparison to current approaches, the suggested solution for real-time smart attendance may offer the following advantages and benefits:

Real-time attendance systems demand quick and effective processing, and the suggested solution might provide quicker processing times than current approaches. The suggested approach might be more scalable and able to manage bigger populations of people or attendance events. The suggested approach may be more tolerant to changes in lighting, position, and facial expressions, enabling more accurate attendance tracking. The suggested solution may include privacy-preserving measures that safeguard people's personal information and shield attendance data from unauthorised access. The suggested solution makes it possible to track attendance in real-time and automatically, doing away with the necessity for manual attendance taking. This not only saves time and labour, but also lowers the risk of fraud or errors compared to manual or paper-based attendance systems.

A camera module, pre-processing module, facial recognition module, database module, and GSM module are all part of the system. Each module is responsible for a specific duty in order for the system to be successfully implemented. The camera module captures photos of faces, which are then processed by the pre-processing module to make them acceptable for facial recognition.

References

1. Alamri, K. A. Amer, A. A. Aldosari, B. M. S. Althubait, M. S. Alqahtani, A. A. M. Al Mudawi, B. A. M Al Mudawi, F. A. M Alqahtani, N. S. Y. Alhamoud, J. Fam. Med. Prim. Care **11**, 9 (2022)
2. K. Wangsan, P. Assavanopakun, P. Upaphong, R. Sapbamrer, W. Sirikul, A. Kitro, N. Sirimaharaj, S. Kuanprasert, M. Saenpo, S. Saetiao, T. Khamphichai, J. Env. Pub. Health, **19**, 7 (2021)
3. G. A. Mersha, M. S. Hussien, G. T. Belete, M. T. Tegene, Hindawi, **2020**, 2561703, (2020)
4. A. K. Lema, E. W. Anbesu, Sage Open Medi. **9**, 20503121221142402, (2021)
5. J. Gomez, J. Coronel Ocampos, P. P. Quiroga Castañeda, A. Gomez, M. J. Valladares-Garrido, Front Public Health **10**, 3389, (2022)
6. F. Adane, Y. M. Alamneh, M. Desta, Tropical Medicine and Health, **50**, 26 (2022).

7. S. Aldokhayel, L. Al Tawil, L. Zeitouni, S. Hussein, T. Qadoumi, S. S. Ahamed, *Europ. J. Ophthal.* **30**, 1 (2020).
8. E. M. Arttime Rios, F. Sanchez Lasheras, A. Suarez Sanchez, F. J. Iglesias Rodriguez, M. D. M. Segui Crespo, *Sensors*, **19**, 2800, (2019)
9. S. H. Venkatesh , A. T Girish, Shashikala , P. Kulkarni, S. Mannava, R. Rajarathnam, *J. Clini. Diagn. Res.*, **11** (2017)
10. Ghufraan A. Abudawood, Heba M. Ashi, Nawaf K. Almarzouki, *J. Ophthal.* **2020**, 2789376, (2020)
11. D. Zenbaba, B. Sahiledengle, M. Bonsa, Y. Tekalegn, J. Azanaw, V. Kumar Chattu, *The Scien. Worl. J.* **2021**, 3384332, (2021)
12. L. Wang, X. Wei, Y. Deng, *J. of Ophth.* **9**, 696036, (2021)
13. M. S. Brau, B. Domenech Amigot, F. Brocal Fernández, J. A. Quesada Rico, M. Seguí-Crespo, *Int. J. Envi. Res. Publ. Heal.*, **17**, 1003, (2020)
14. A. Das, S. Shah, T. B. Adhikari, B. S. Paudel, S. K. Sah, R. K. Das, C. P. Shah, P. G. Adhikari, *PLoS One*, **5**, 8, e0236766, (2020)
15. B. S. Kumar, *Biomed. Pharm. J* **13**, (2020)
16. R. P. Ram Kumar, P. Sanjeeva, S. F. Lazarus, D. V. Krishna, *Intl. J. Inno. Tech. Explor. Engg* **8**, 11S2 (2019)
17. D. P. Degala, M. Anjaneyulu, P. Devika, *Ergonomically Designed System for License Plate Recognition Using Image Processing Technique*, in *Proceedings of the Confidential Computing. Advanced Technologies and Societal Change*, Garcia Diaz, , Rincón Aponte, (eds), Springer, Singapore, (2022)
18. B. Srikanth, M. Santoshi Kumari, C. Sravani, S. G. Bhargavi Latha, M. Suresh, *Human Activity Identification using Artificial Intelligence*, *Research Advances in Intelligent Computing*, 1st Edition, CRC Press (2023)
19. R. N. Ashlin Deepa, P. Karlapati, M. R. Mulagondla, P. Amaranayani, A. P. Toram, *An Innovative Emotion Recognition and Solution Recommendation Chatbot*, in *Proceedings of the 2022 8th International Conference on Advanced Computing and Communication Systems (ICACCS)*, Coimbatore, India, (2022)